PINN design: architecture and inductive biases

The observability problem

- Surveillance reports only two series: cumulative cases C and deaths D
- Hidden compartments S, E, I are **never measured**
- Transmission $\beta(t)$ and mortality μ must be inferred from C and D alone

Observable re-parameterisation

Set $C = I + R + D$ (all symptomatic). Then $S = N - E - C$ exactly, so total population conservation $S+E+I+R+D = N$ holds identically for any network output.

Dual-network architecture

- **State net** $g_\theta(\tau)$: 3×48 tanh units, 4 positive outputs:

$$e = e_0\, e^{\tau}\, g_1, \quad i = i_0\, e^{\tau}\, g_2, \quad c = c_0\, e^{\tau}\, g_3, \quad d = d_0 + \tau\, g_4$$

- **Transmission net** $h_\varphi(\tau)$: 3×32 tanh units:

$$\beta(\tau) = \mathrm{Softplus}(h_\varphi(\tau) + b) > 0$$

Six inductive biases

1. **Hard ICs:** Exponential factor $e_0 e^{\tau} g_1$ guarantees $e(0)=e_0$ exactly; τ factor enforces zero IC gradient at $\tau=0$.
2. **Positivity:** Softplus and exp mappings strictly prevent negative populations.
3. **CFR bounds:** $\mu = \gamma\,\mathrm{CFR} / (1-\mathrm{CFR})$ with $\mathrm{CFR} \in (0.2, 0.8)$, eliminating the degenerate $\mu \to 0$ attractor.
4. **Fourier features:** Inputs $[\sin(k\pi\tau), \cos(k\pi\tau)]$ for $k=1..4$ defeat neural spectral bias so rapid $\beta(t)$ shifts are captured.
5. **Relative ODE residuals:** Each residual scaled by state magnitude so early small-scale dynamics are weighted equally with the peak.
6. **Smooth prior on β:** Penalty on $(d\beta/d\tau)^2$ renders the ill-posed inverse problem well-conditioned.

What the architecture guarantees without training: Total population conservation $S+E+I+R+D = N$ to machine precision (2×10^{-16}); positive states; exact initial conditions; biological CFR range [20%, 80%].

Loss function and training protocol

Total loss formulation

$$L = (L_C + L_D) + L_{ode} + w_{sm} L_{smooth}$$

- L_C, L_D: Mean squared error evaluated exclusively at real report dates (no interpolation artifacts).
- L_{ode}: Autograd relative ODE residuals over $N_f = 200$ collocation points:

$$\Sigma\,[(e^t - Tf_e)/e]^2 + [(i^t - Tf_i)/i]^2 + [(c^t - Tf_c)/c]^2 + [(d^t - Tf_d)/d]^2$$

- L_{smooth}: Mean squared derivative $\Sigma\,(d\beta/dt)^2$.

Why relative ODE residuals?

An outbreak grows exponentially across orders of magnitude. Under standard absolute residuals, early-stage errors contribute negligible loss. The E6 ablation reveals absolute residuals yield **75%–128% error** vs. < **0.2%** under relative scaling.

Two-stage optimiser

- **Stage 1 — Adam:** LR 2×10^{-3} with cosine annealing, 2000 iterations. Resamples random collocation points per step to escape local minima.
- **Stage 2 — L-BFGS:** Strong-Wolfe line search (2500–6000 iters) on a fixed linspace grid for quadratic convergence.
- **Float64 precision:** Strict double precision prevents rounding errors during higher-order autograd differentiation.
- **Ensemble:** 5 random seeds; median trajectory acts as robust point forecast.

Forecasting protocol

Hold β at its empirical 14-day trailing mean; freeze inferred μ; propagate SEIRD forward via DOP853 numerical solver.

Ablation insights (E6)

Adam-only: *e(t)* error jumps to **96%**. Physics weight $w_{ode} = 10$: errors triple. Fourier features reduce β RMSE by **28%**.

Key design decision: The L-BFGS second-order refinement is indispensable. Adam alone establishes coarse trends, but fails to resolve stiff ODE phase trajectories.

Synthetic experiments: E2 forward, E3 inverse β(t), E4 hidden states

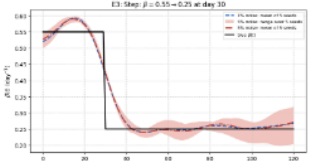
E2: Forward solution verification

- Trained solely on physics loss L_{ode} with **no data** ($w_{data} = 0$)
- Pointwise errors against DOP853 < $\mathbf{2.0\times10^{-3}}$ across all states
- Total mass conservation error: $\mathbf{2\times10^{-16}}$ (machine zero)

E3 & E4: Quantitative recovery under noise

Target Quantity	1% Noise	5% Noise
β(t) relative L_2	11.0% ± 0.7%	11.6% ± 0.6%
β(t) RMSE (>5d from jump)	0.022 ± 0.005	0.025 ± 0.005
μ relative error	4.6% ± 0.3%	5.4% ± 1.0%
Hidden s(t)	0.36% ± 0.03%	0.46% ± 0.09%
Hidden e(t)	5.8% ± 0.6%	6.4% ± 0.7%
Hidden i(t)	5.8% ± 0.4%	6.5% ± 0.8%

Results: mean ± SD over 5 seeds; step reduction β: 0.55 → 0.25 at Day 30.



E2 Forward: Worst-case compartment error < 2.0×10^{-3}; mass conservation error 2×10^{-16}. **E3 Inverse:** Recovers non-smooth β(t) within 11.6% under 5% noise. **E4 Hidden states:** Reconstructs susceptible curve to 0.46% with *zero direct measurements*.